



Create Your Own Java Based Chat Robot

If you are interested in creating your own chat robot, then here is a tutorial on how to do it yourself by using Java and AIML.

A chat robot or chatterbot is a human chat simulator. It is a program for auditory or textual conversation between a computer and a human being. Such robots are used for fun, education and 24x7 customer services. When students and customers have many questions to ask, the robot gives the answers on behalf of teachers or customer service executives.

In this article, I have used AIML, A.L.I.C.E, Java and NetBeans. A brief introduction to these technologies follows.

AIML (Artificial Intelligent Markup Language) is an XML based mark-up language to help create a chat robot from scratch. It was first developed by Dr Richard Wallace when he created a chat robot named A.L.I.C.E (Artificial Linguistic Internet Computer Entity).

The important tags for AIML are listed below.

- `<aiml>`: The parent tag to start and end the AIML document.
- `<category>`: Every new question or pattern with its relevant answer goes in this tag.
- `<pattern>`: Matches the pattern with the user's question.
- `<template>`: The robot gives the answer from the

template if the pattern can be matched.

```
<?xml version = "1.0" encoding = "UTF-8" ?>
<aiml version = "1.0.1" encoding = "UTF-8"?>
  <category>
    <pattern> what is your name? </pattern>
    <template>
      I am Alice, nice to meet you.
    </template>
  </category>
</aiml>
```

Here is an explanation of the code given above. File *first.aiml* contains the XML based tag for the robot's knowledge. Here, the `<category>` tag is used to describe the user pattern or the user's question. `<template>` is the response given by the robot to the user if the user's pattern is matched.

A.L.I.C.E is the robot created in 1995 by Dr Richard Wallace, in Java, by using AIML. We can also say that A.L.I.C.E is an AIML parser. Nowadays, many parsers are available in various languages like PHP, Python, etc.